

Smart Parking System — Testing Report

Testing objective

Verify the four-slot occupancy logic, available count, OLED, LEDs, buzzer, servo demonstration, and serial output in Wokwi.

Test matrix

ID	S1	S2	S3	S4	Expected Free	Expected Result
TC01	100	100	100	100	4	All FREE
TC02	10	100	100	100	3	S1 OCCUPIED
TC03	10	15	100	100	2	S1,S2 OCCUPIED
TC04	10	15	20	100	1	S1,S2,S3 OCCUPIED
TC05	10	15	20	12	0	PARKING FULL + buzzer
TC06	10	100	20	12	1	Vehicle departure/recovery

Evidence checklist

- 1 01_initial_circuit.png — complete connected circuit
- 2 02_empty_parking.png — all four slots free
- 3 03_slot1_occupied.png — one occupied
- 4 04_two_slots_occupied.png — two occupied
- 5 05_three_slots_occupied.png — three occupied
- 6 06_parking_full.png — all occupied, buzzer and full state
- 7 07_vehicle_departure.png — one slot freed
- 8 08_oled_status.png — readable OLED
- 9 09_led_status.png — mixed green/red indication
- 10 10_buzzer_alert.png — full alert
- 11 11_servo_gate.png — gate demonstration
- 12 12_serial_monitor.png — diagnostic output
- 13 13_final_simulation.png — integrated final state

Pass/Fail policy

A test is PASS only when the observed Wokwi output matches the expected behavior. The package intentionally leaves actual results blank so the evidence remains truthful.

Calibration notes

Start at 35 cm. If required, adjust by approximately ± 5 cm based on sensor placement and verify that averaged readings are stable. The implementation uses five readings with min/max trimming and 800 ms debounce.

Final sign-off

Tester: _____ Date: _____

Final result: _____